EXPRIMENT-18

18.) DES Subkey Generation

Aim:

To write a Python program to demonstrate the generation of a 48-bit DES subkey using two 28-bit halves.

PROGRAM:

```python
def generate_subkey(c, d):

    left = c[:24]
    right = d[:24]

    subkey = left + right

    return subkey


c = "10101010101010101010101010"
d = "01010101010101010101010101"

key = generate_subkey(c, d)

print("First 24 Bits:", key[:24])
print("Second 24 Bits:", key[24:])
print("Generated Subkey:", key)
print("Subkey Length:", len(key))
```

OUTPUT:

```
First 24 Bits: 101010101010101010101010
Second 24 Bits: 010101010101010101010101
Generated Subkey: 101010101010101010101010010101010101010101010101
Subkey Length: 48
```

RESULT:

Thus, a 48-bit subkey was generated using two separate 28-bit halves.